

PRATIK SHRIVASTAVA

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PROFESSIONAL SUMMARY

Aspiring Software Engineer with solid foundation in programming, Data Structures, Algorithms, Databases, and backend development. Experienced in building real world applications using Python, JavaScript, FastAPI, SQL and Git. Eager to leverage technical and analytical skills to develop software solutions and deliver business impact.

WORK EXPERIENCE

Data Science Intern

Jun 2025 – Jul 2025

Dodge AI

Remote – Bengaluru, KA

- Designed ML pipeline predicting customer purchase intent, benchmarking 5 classifiers and deploying a Random Forest model via Flask.
- Built a hybrid recommendation engine combining TF-IDF filtering with SVD-based collaborative filtering on personalized product recommendations.
- Conducted EDA and engineered behavioural features, applying SMOTE to handle class imbalance and improve prediction accuracy.

Software Engineering Intern

May 2024 – Aug 2024

Siemens

Remote – Bengaluru, KA

- Developed a near real-time frontend dashboard for an IIoT-based water monitoring platform, visualizing live consumption data from distributed sensors through interactive dashboards.
- Integrated RESTful APIs and implemented near real-time data updates, enabling operational monitoring and historical consumption trend analysis.
- Built GIS-based map-based visualizations with live meter status indicators and alerts, improving visibility into device health and flow anomalies across multiple locations.

PROJECTS

Drape | 🌐 [Website](#)

- Built an AI-powered visual fashion search engine that finds visually similar catalogue products from an uploaded photo, using YOLOv8 for garment detection, DINOv2 (768v) for vector embeddings, and FAISS for similarity search.
- Designed a neighbour-consensus that derives garment descriptions and retailer search links directly from retrieval results – avoiding a second model on constrained memory budget by making descriptions a direct readout of retrieval quality.
- Deployed FastAPI backend on AWS Lambda as a container image, serving ~ 1-2s warm inference over a 20k-image catalogue within free tier limits.

Harmoniq | 🌐 [Website](#)

- Developed a hybrid recommender blending 4 retrieval sources – latent KNN embeddings, raw audio features, genre families, and popularity into one ranked score, with a per-result panel explaining each signal's contribution.
- Deployed the model (FastAPI on Render, React/Vite on Vercel) used Spotify OAuth, also cold cutting start latency by preloading artifacts on startup using a scheduled CRON job at a 10-minute interval.
- Shipped surfaces on one artifact set – recommendations, mood discovery, genre exploration, playlist building, 3D PCA / feature correlation visualizations, catalogue overview and a methodology page.

Booktures | 🌐 [Website](#)

- Built an AI-powered book illustration pipeline that extracts texts from PDFs via OCR, identifies characters using configurable NER (spaCy or a single LLM pass per book) with union find alias merging, and generates page-consistent illustrations by feeding a running visual profile in each prompt.
- Designed a layer supporting multiple LLMs and image generation APIs including Groq, Gemini and Cloudflare Workers AI.
- Deployed and maintained application on Render, Vercel and Supabase PostgreSQL with a support of running the whole application locally using HF Diffusers and Ollama models.

PROJECTS

Programming Languages	Python, C++, JavaScript, SQL
Frameworks & Libraries	Express, Node.js, Fast API
Tools & Platforms	Github, Git, Postman
Databases	MySQL, PostgreSQL, Supabase
ML & Data	Scikit-learn, Transformers, Pandas, Numpy, OpenCV, Keras

EDUCATION

Manipal University Jaipur

July 2022 – July 2026

Bachelor of Technology in Computer Science & Engineering